

Global CO₂ Emissions

Who Emits, How Much, and What's Changing

Scope: Fossil fuels & industry, 1750–2024

Data Attribution

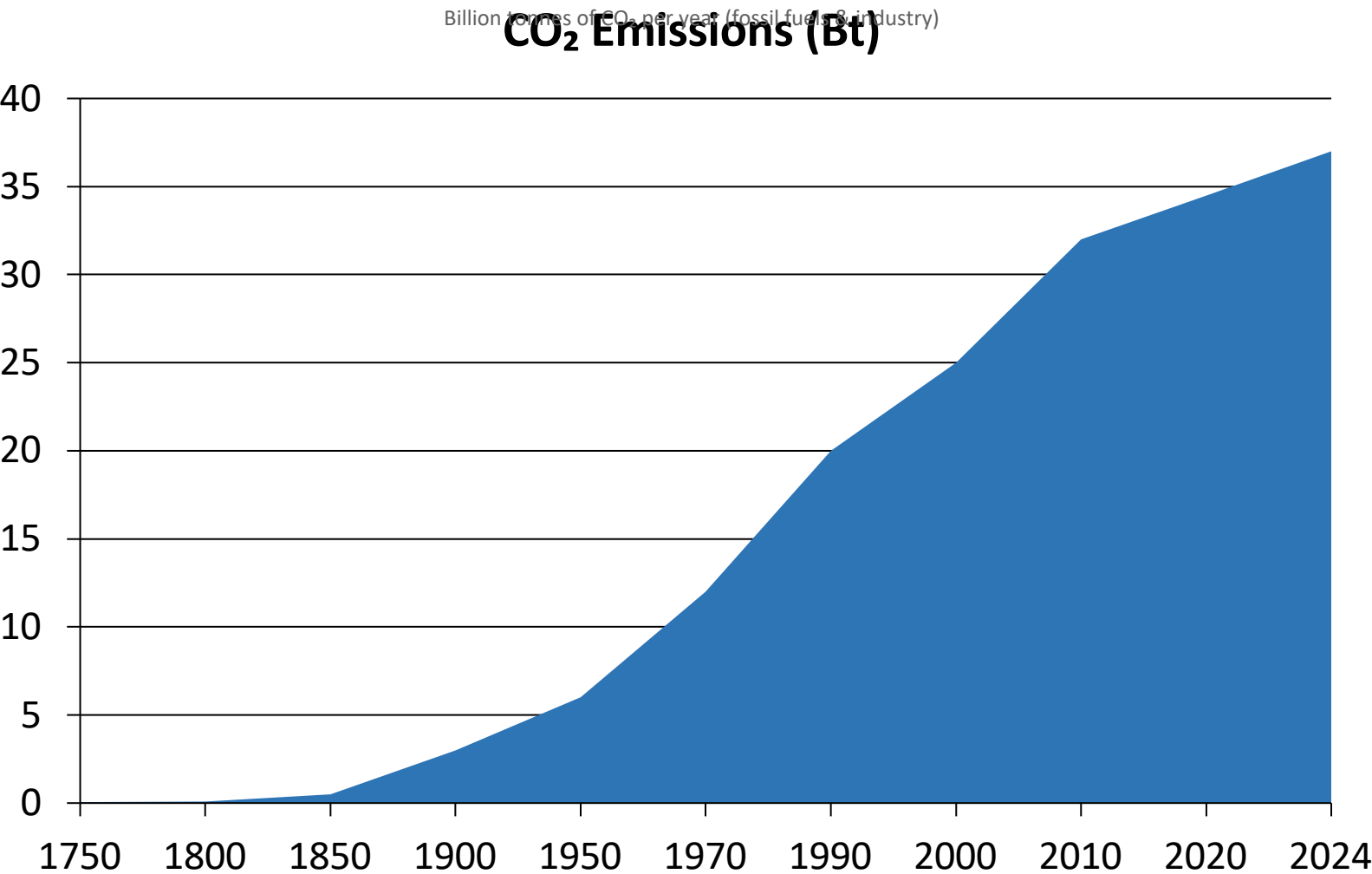
Global Carbon Budget 2025 via Our World in Data

Authors: Hannah Ritchie and Max Roser | ourworldindata.org/co2-emissions

Audience

Policy analysts, climate negotiators, sustainability researchers

Global CO₂ Emissions Grew from 6 Bt in 1950 to Over 35 Bt Today



Key Inflection Points

Pre-1750

Negligible emissions from biomass & land use

1750–1850

Slow rise from early industrialization (UK, Europe)

1850–1950

Accelerating growth; coal-powered industrialization

1950–1990

Rapid doubling; globalization & oil era

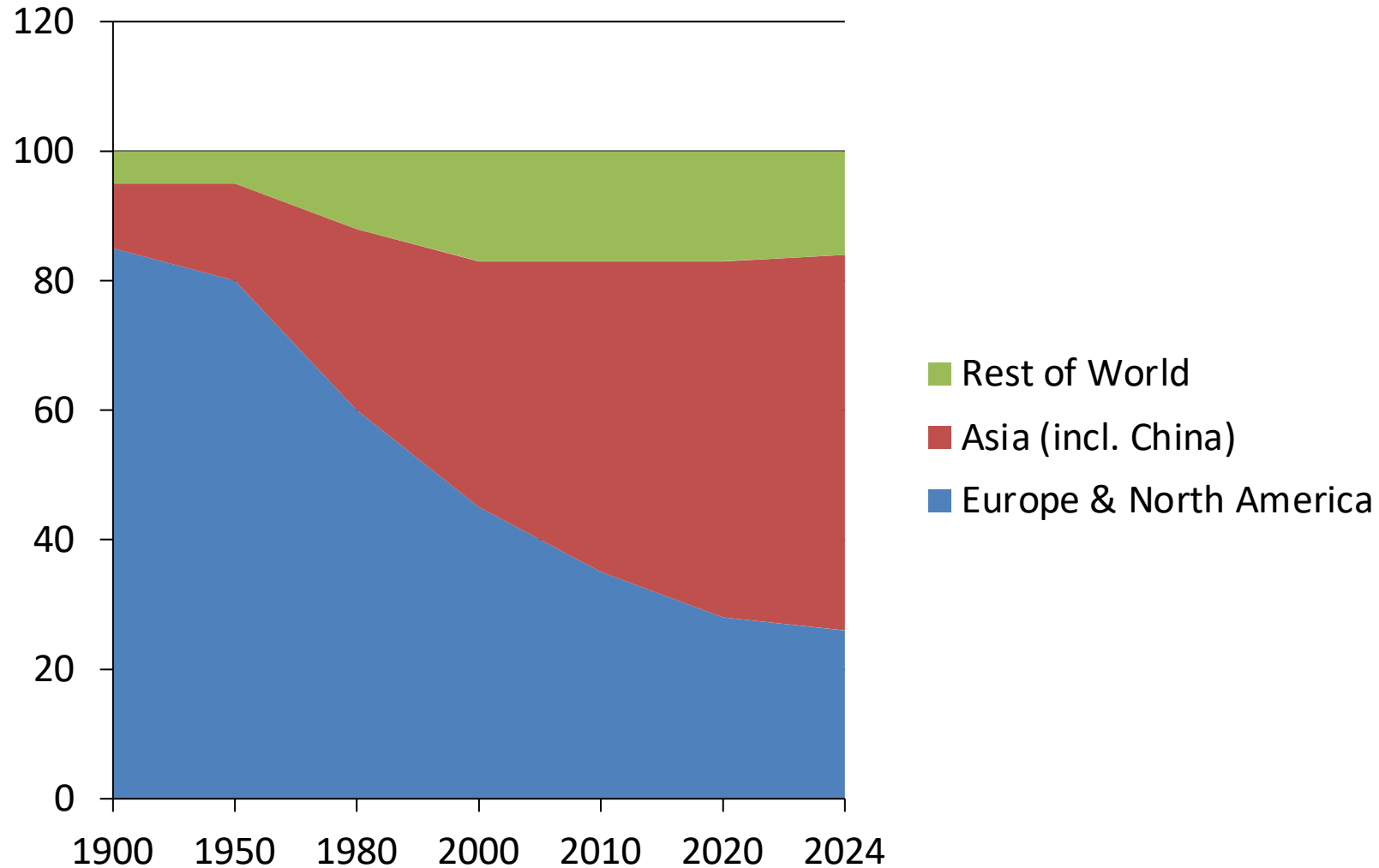
1990–2024

Growth slowing but no peak yet; China surge

2024

~37 Bt — highest ever recorded

Europe and the US Fell from 90% of Emissions in 1900 to Under One-Third Today



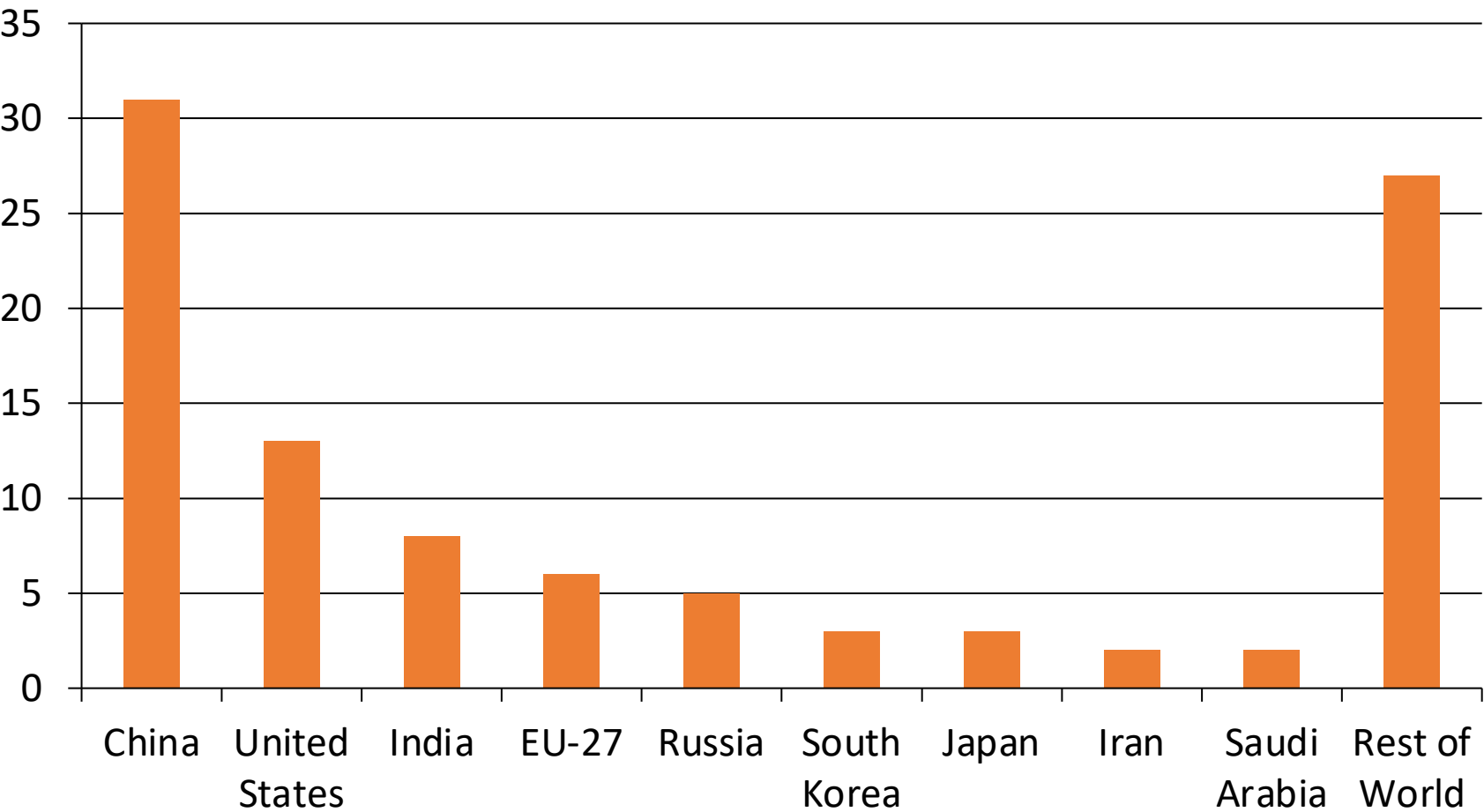
The Great Shift

- In 1900: Europe + US dominated at 85%+ share
- By 1950: Still 80% combined despite rise of US industry
- By 2024: Fallen to ~26% combined
- Asia now produces ~58% of global emissions — up from just ~10% in 1900
- China alone accounts for ~31% in 2024

Chart: Stacked area — share of global emissions by region

China Now Emits More Than One-Quarter of Global CO₂

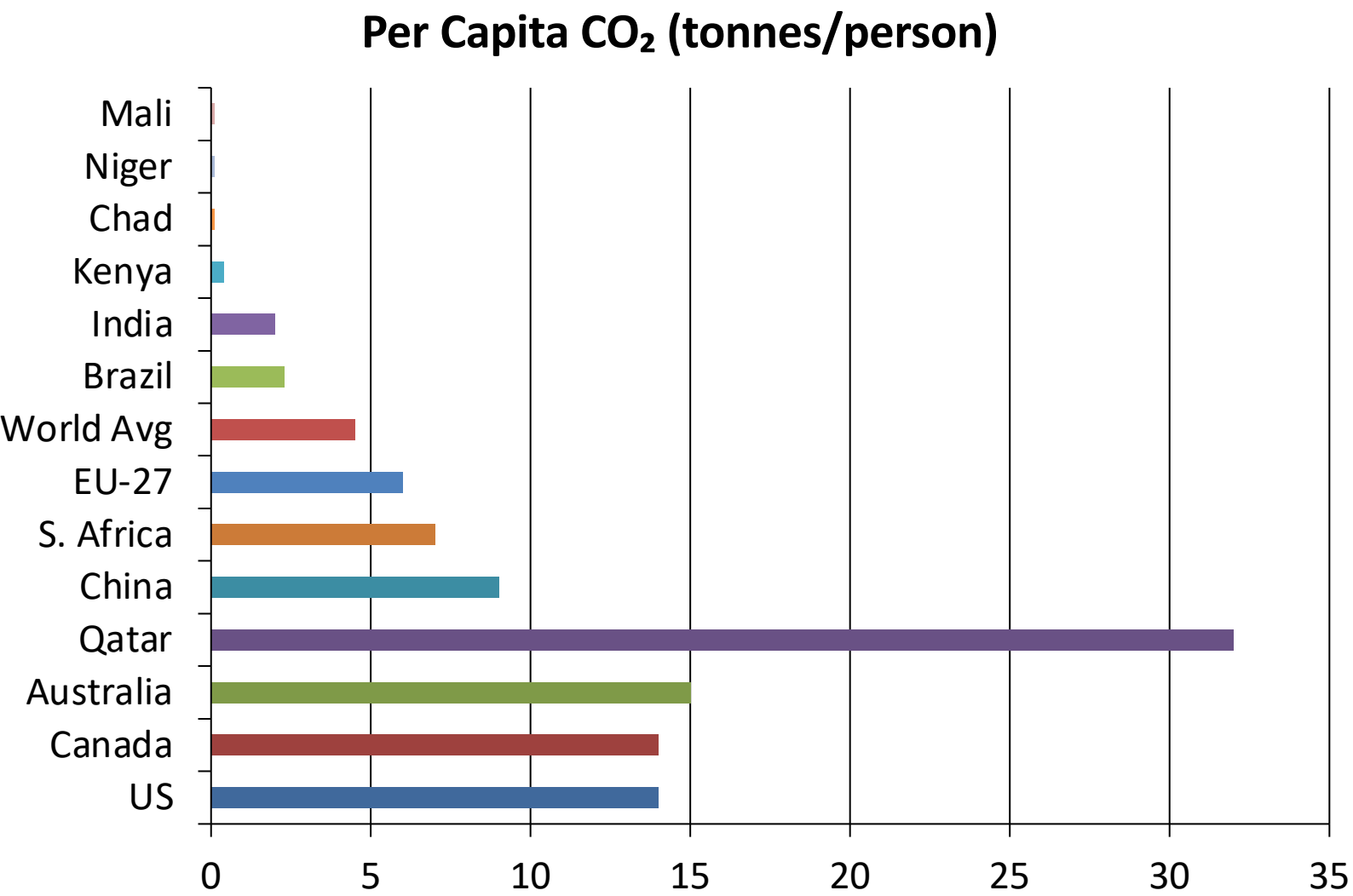
Share of Global CO₂ (%)



2024 Country Shares		
China		31%
United States		13%
India		8%
EU-27		6%
Russia		5%
Rest of World		27%
Int'l Aviation & Shipping		~3%

Source: Global Carbon Budget 2025 via Our World in Data | ourworldindata.org

Per Capita Emissions Range from 0.1 t in Chad to ~14 t in the US and Canada



The 150x Gap

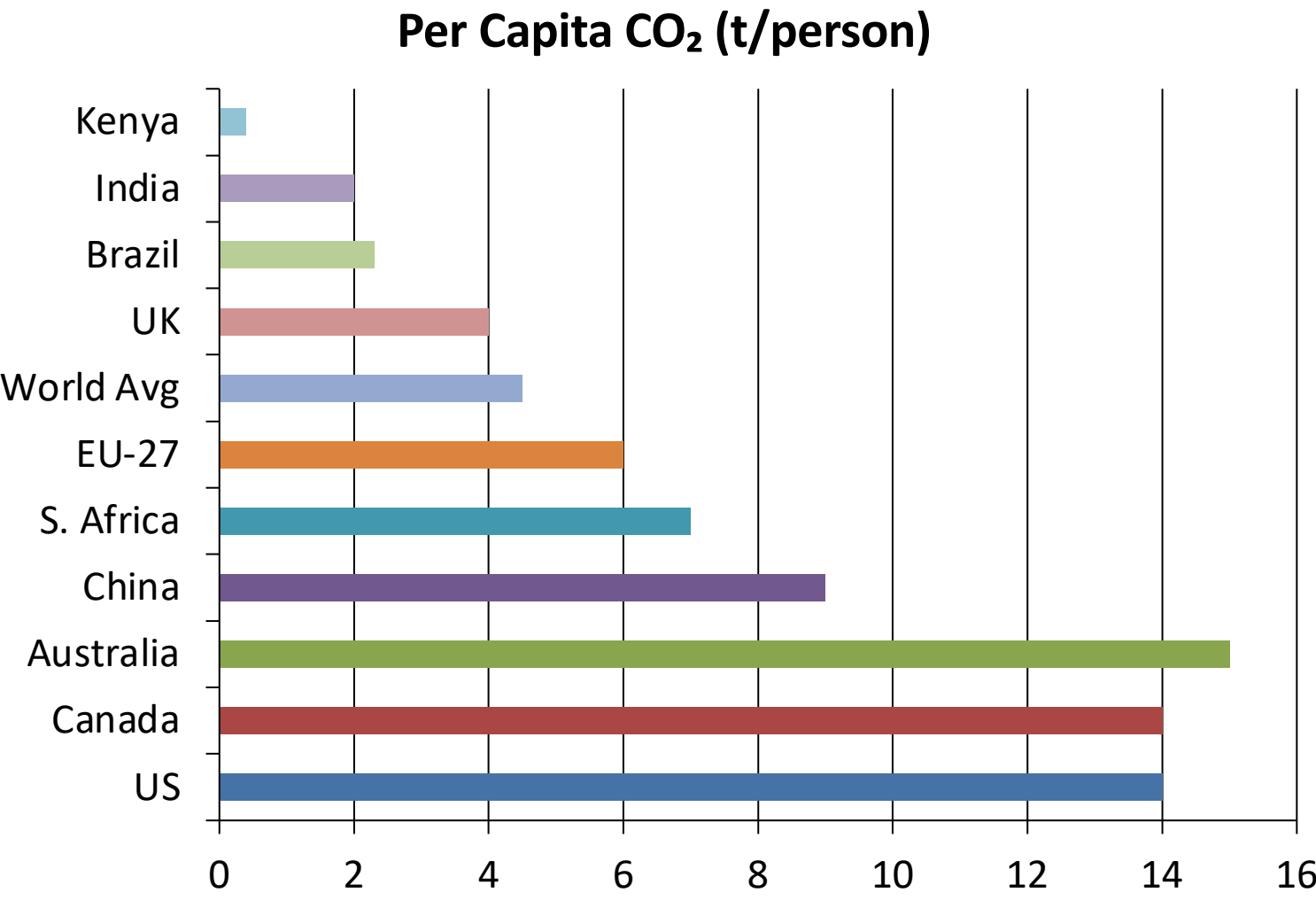
The difference between the highest and lowest emitting nations represents a ~150x inequality.

An American emits in under 2 days what a person in Chad, Niger, or Mali emits in an entire year.

Note:

- Qatar (32 t) — small oil-rich nation; outlier
- France & UK (~4 t) — wealthy but low per capita due to nuclear/renewables
- India (~2 t) — large population, low per capita despite rapid growth
- Sub-Saharan Africa — lowest per capita, least responsible for historical emissions

Per Capita Emissions Vary Widely: ~14 t in the US vs. ~9 t in China and ~2 t in India



Key Observations

China (~9 t)

Largest total emitter, but per capita less than 1/3 of US — and still below the world average when divided by 1.4B people

India (~2 t)

Per capita less than 1/5 of US; 1.4B people with low emissions growth trajectory

EU-27 (~6 t)

Wealthy region with lower per capita than US due to nuclear & renewable energy mix

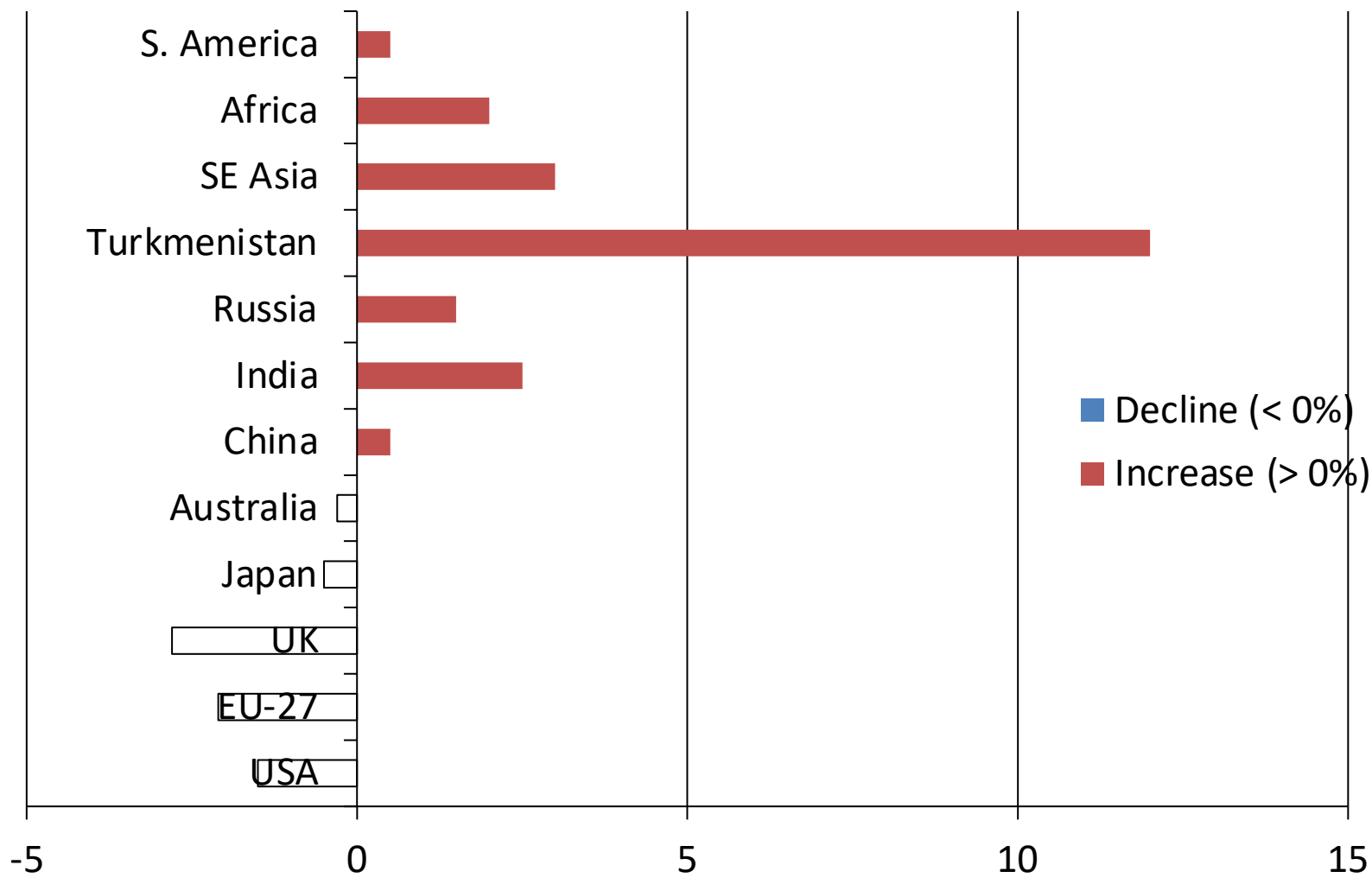
UK (~4 t)

Similar living standards to US but less than 1/3 the per capita emissions

World Avg (~4.5 t)

Global average masks enormous within-country inequality

2024 Snapshot: Many Western Nations Cut Emissions While Parts of Asia and Africa Grew



2024 Regional Changes

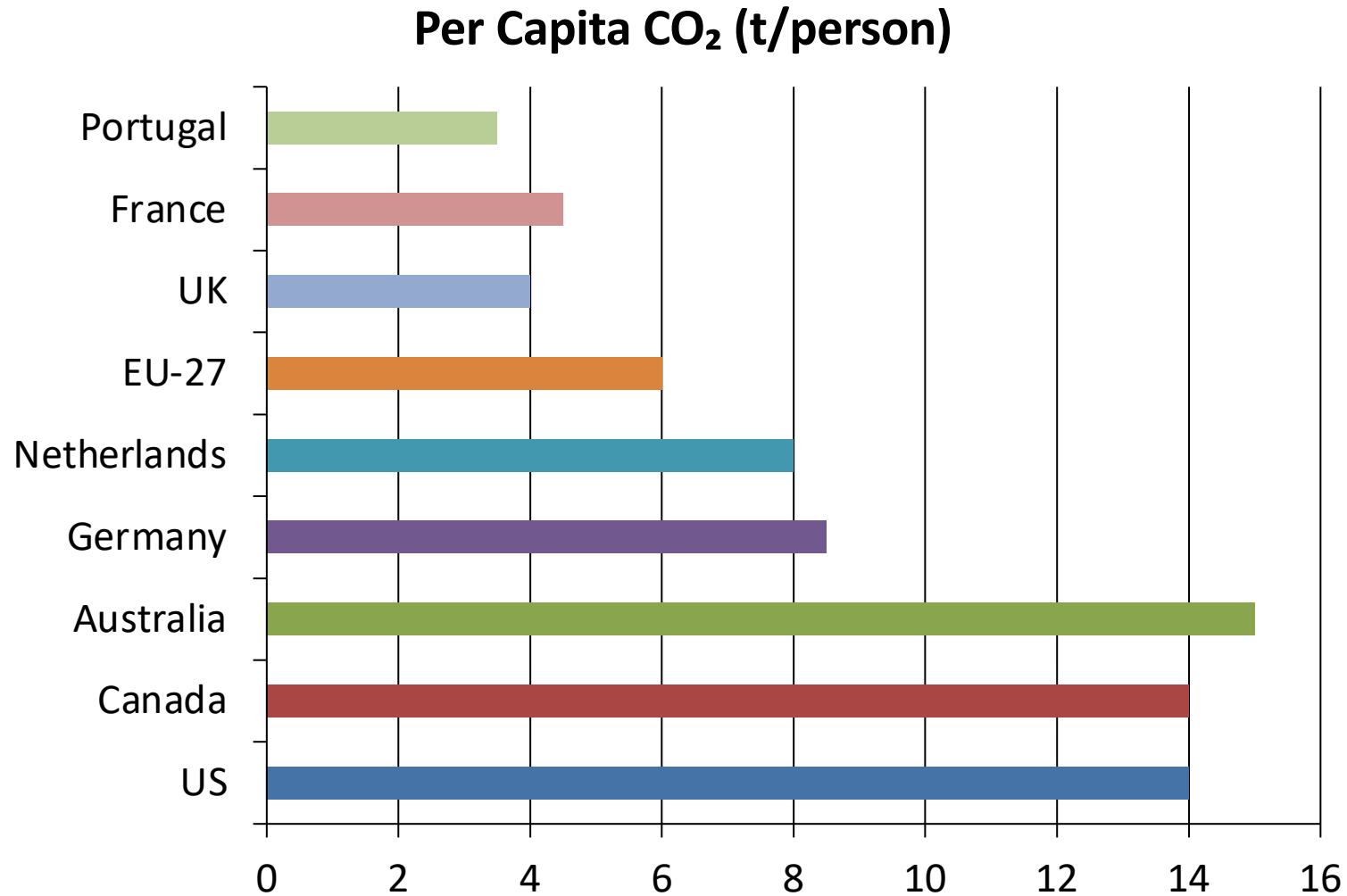
- Decline: US, EU, UK, Japan, Australia
- Increase: Russia, India, SE Asia, Africa

Notable outliers

Turkmenistan/Uzbekistan: >10% increase (largest gains globally)
India: ~2.5% increase — growing emissions from industrialization
China: ~0.5% — growth slowing significantly

Chart: Diverging bar showing % change in CO₂ emissions, 2024

Energy Mix Explains Why France and the UK Emit Far Less Per Capita Than Germany or the US



Why Energy Mix Matters

Same prosperity, different emissions

France and the UK have similar living standards to Germany and the US, yet emit 2–3× less per capita.

Nuclear power is key

France (~75% nuclear electricity) achieves ~4.5 t/capita vs. Germany's ~8.5 t — even though both are wealthy industrial nations.

Renewables growing fast

Portugal's grid is increasingly wind and solar, driving its ~3.5 t/capita vs. Netherlands' ~8 t.

Policy decoupling

These examples prove that high-income countries can maintain prosperity while dramatically reducing emissions through energy choices.

Key Takeaways: Responsibility Is Shared but Unequal

1 35 Bt and rising — no peak yet

Global fossil fuel CO₂ emissions exceed 35 billion tonnes per year and have not peaked. Despite slowing growth rates, emissions continue to climb.

2 China dominates total; per capita still below US

China (~31%) is the world's largest annual emitter. Yet its per capita emissions (~9 t) remain less than ⅓ of the US (~14 t), reflecting its large population and manufacturing role in the global economy.

3 Historical responsibility: Europe & US

Europe and North America once produced >90% of global emissions. Today they produce less than one-third — but their cumulative historical emissions are the primary driver of atmospheric CO₂ accumulation.

4 150x per capita gap between richest and poorest

The difference between top emitters (US, Canada, Australia: ~14–15 t/person) and bottom emitters (Chad, Niger, Mali: ~0.1 t/person) represents the starkest dimension of global inequality.

5 Energy policy can decouple prosperity from emissions

France (~4.5 t), the UK (~4 t), and Portugal (~3.5 t) show that wealthy nations can maintain high living standards with far lower per capita emissions through nuclear power and renewable energy.